1) What are the absolute minimum dimensions of a transistor ? Explain your reasoning. Remember to consider minimum contact size. Do we actually use this minimum size or do we use a slightly larger size for convenience.

I feel like, there is no single universal absolute minimum, I think so finfet technology would be the ~~minimum~~ technology with min dimensions.
In our course work, we use 180 nm technology,

NMOS :-

$$A_S = A_d = 5\lambda W \qquad P_S = P_d = 10\lambda + W$$

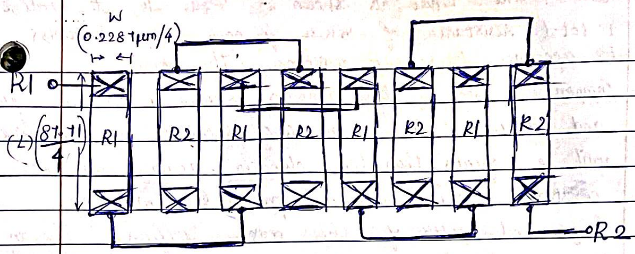
PMOS :-

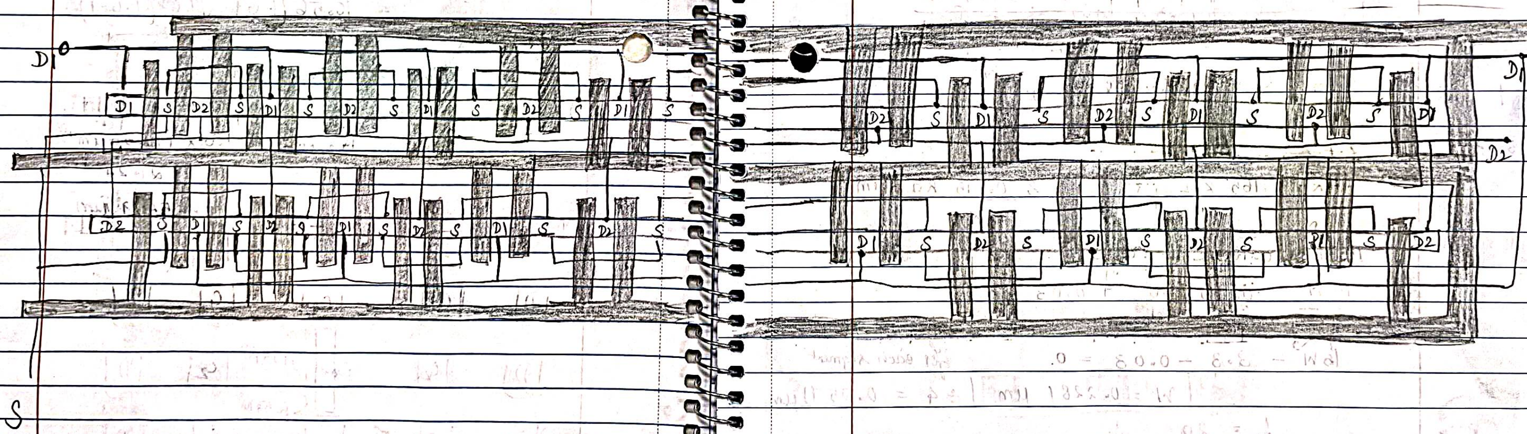
$$A_S = A_d = 5\lambda W \qquad P_S = P_d = 10\lambda + 1.5W.$$

Slightly larger size is preferred for convenience
Inorder to satisfy, minimum contact size, required enclosures and spacings.

2) Draw a common-centroid layout of a simple current-mirror with equal size transistors of $L = 180$ nm and $W = 12\,\mu m$. Use a finger width of 500 nm for each transistor segment. Remember to draw the drain, source, gate and bulk connections. Use dummy transistors and all other good layout techniques learned in the lab. Include a floor plan

$$\text{no. of fingers} = \frac{12\,\mu m}{0.5\,\mu m} = 24 \text{ fingers}$$





3, $C_1 : C_2 = 1 : 1.4$

$D_1 = D_2 = 4.$

Unit Capacitor = $10.56\,\mu m \times 10.56\,\mu m$. $(W \times L)$

$$\frac{1}{1.4} = \frac{4C_u}{4C_u + NC_u}$$

$$4C_u + NC_u = 5.6\,C_u.$$

$$4 + N = 5.6$$

$$N = 1.6.$$

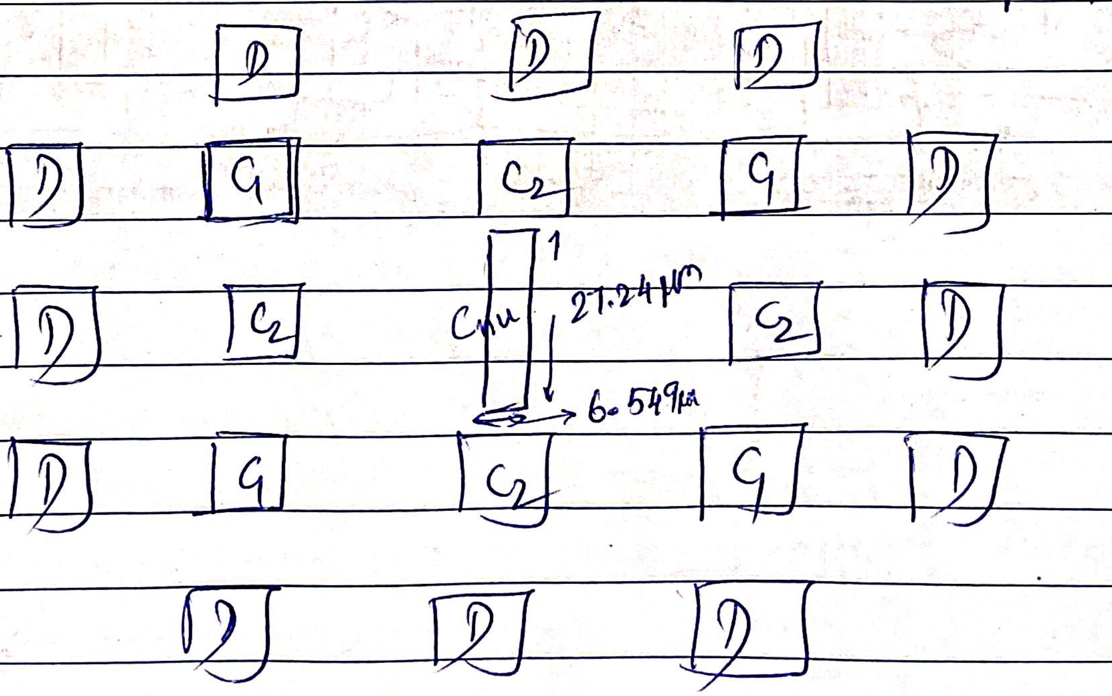
$$L_{nu} = \cancel{10.}\ L_0\left(N + \sqrt{N(N-1)}\right)$$

$$= 10.56\left(1.6 + \sqrt{(1.6)(1.6-1)}\right)$$

$$L_{nu} = 27.24\ \mu m.$$

$$W_{nu} = \frac{N \times L_0^2}{L_{nu}} = \frac{1.6 \times (10.56)^2\,\mu m^2}{27.24}$$

$$= 6.5499\ \mu m$$



4. Draw a layout for two matched polysilicon resistors, like what is shown in Figure 3.14, to realize a total resistance of 15 KΩ for each resistor. Remember to account for contact resistance. The total component area of both resistors should be between 20 μm^2 and 80 μm^2 (the actual area occupied by the layout will be greater than this due to routing and DRC constraints like minimum spacings. Determine the length and width of the resistor diffusion. Use an interdigitized layout with four resistance segments for each resistor. Use good layout techniques.

Answer :-

Let the total area be = 40 μm^2.

Area of 2 resistors = 20 μm^2.

$$< 20\,\mu m^2 = W \times L. \quad \text{—} (1)$$

$$L = \frac{20\,\mu m^2}{W}$$

$$R_{nom} = R_S \times \frac{L}{W} + 2\frac{R_{con}}{W}$$

8

$\cancel{0.165}$ =

$$15k = 0.165\ k\Omega/sq \times \frac{L}{W} + 2\frac{0.015\ k\Omega - \mu m}{W}$$

$$15W = 0.165L + 0.03$$

$$15W = 0.165 \times \frac{20}{W} + 0.03$$

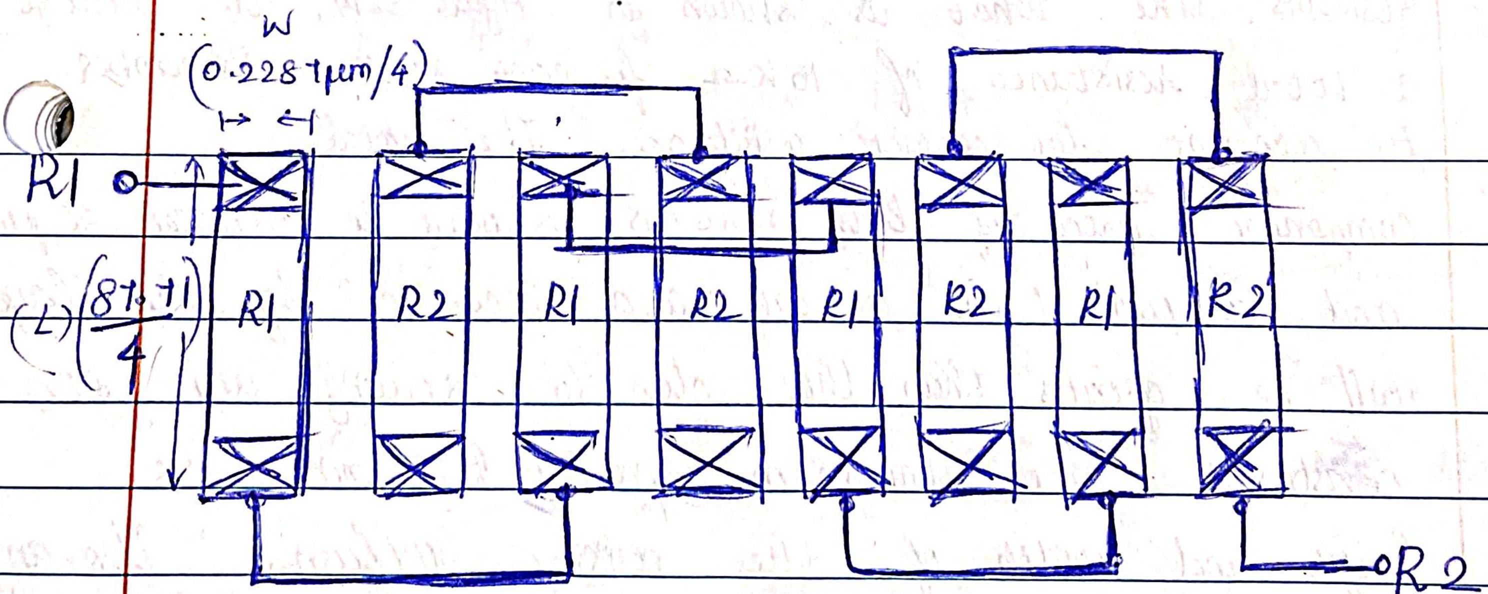
$$15W^2 - 3.3 - 0.03 = 0.$$

for each segment

$$\boxed{W = 0.2287\ \mu m} \div 4 = 0.0571\ \mu m$$

$$L = \frac{20}{0.228}$$

$$\boxed{L = 87.71\ \mu m} = 21.9275\ \mu m$$



W
(0.2287μm/4)
R1
(L) (87.71/4)
R1
R2
R1
R2
R1
R2
R1
R2
R2